STACKS, DISCRIMINANTS: LEMMA-NOETHER-DIFFERENT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1524-1529 of sources/stacks-discriminant/pinned-original-source.tex; proof span is lines 1531-1553, the complete author proof (21 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 27 of queue_discriminant.json (pb=1008, stmt 1524-1529, proof 1531-1553).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 00UT: sources/stacks-discriminant/pinned-original-source.tex, source lines 1553–1553. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00DV: sources/stacks-discriminant/pinned-original-source.tex, source lines 1553–1553. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00RW: sources/stacks-discriminant/pinned-original-source.tex, source lines 1553–1553. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BVN: sources/stacks-discriminant/pinned-original-source.tex, source lines 1335–1345. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $A \to B$ be a finite type ring map. Let $\mathfrak{D} \subset B$ be the Noether different. Then $V(\mathfrak{D})$ is the set of primes $\mathfrak{q} \subset B$ such that $A \to B$ is not unramified at $\mathfrak{q}$.*

Proof. Assume $A \to B$ is unramified at $\mathfrak{q}$. After replacing B by B_g for some $g \in B$, $g \notin \mathfrak{q}$ we may assume $A \to B$ is unramified (Algebra, Definition 00UT and Lemma 0BVN). In this case $\Omega_{B/A} = 0$. Hence if $I = \mathrm{Ker}(B \otimes_A B \to B)$, then $I/I^2 = 0$ by Algebra, Lemma 00RW. Since $A \to B$ is of finite type, we see that I is finitely

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generated. Hence by Nakayama's lemma (Algebra, Lemma 00DV) there exists an element of the form $1 + i$ annihilating I. It follows that $\mathfrak{D} = B$.

Conversely, assume that $\mathfrak{D} \not\subset \mathfrak{q}$. Then after replacing B by a principal localization as above we may assume $\mathfrak{D} = B$. This means there exists an element of the form $1 + i$ in the annihilator of I. Conversely this implies that $I/I^2 = \Omega_{B/A}$ is zero and we conclude. $\square$